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ASSIGNMENT COVER SHEET

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LECTURER NAMES	Mr B Khoza	
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STUDENT NAMES	STUDENT NO	SIGNATURE
Qobo Duke Bhongo	202329998	
Khumalo A	230014698	
Zungu N	202375568	
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Abstract

This project designs a smart airport luggage sorting system to improve efficiency and reduce lost baggage. Two systems, the carousel sorter and the cross-belt sorter, were compared using RFID tracking and SMS notification. After evaluation, the carousel sorter was selected because it is simpler, cheaper, and more reliable for developing environments.

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1. Introduction

Airports manage many passengers and luggage every day, making baggage handling an important part of operations. An inefficient system can lead to delays, lost luggage, and poor customer satisfaction. These challenges are more serious in developing countries where resources and infrastructure may be limited. This project aims to design a smart and reliable luggage sorting system. It uses RFID technology to track bags and SMS notifications to update passengers. Two sorting systems are evaluated to find the best solution that balances cost, efficiency, and ease of maintenance.

2. STEEP Analysis

2.1 Social

1. This specific design project tests the final year engineering students' skills and knowledge transfer.
2. Reduces the risk of incidents because the design will reduce manual lifting, pushing, and handling of baggage, since annually baggage handling injuries cost a lot of money to airports [1]
3. Workers will not be happy with the stakeholders, and the engineering designers (Bhongo, Andisa, and Ntethelelo) as automatic systems are introduced, and so we need to unveil a series of measures to balance automatic and manual systems. [1]
4. The community may view the system as effective due to the improved efficiency

2.2 Technological

1. The main problem is the infrastructure in developing countries like South Africa. Power and the internet are not always stable; sometimes they go off. Because of this issue, the system must use local processing through a PLC or edge controller so it can work even when the network is down [2].
2. Integration is also not easy because many airports still use the old barcode system. So instead of replacing everything, it is better to use both RFID and barcode together so the new system can work with the existing airport barcode infrastructure while slowly adding RFID tracking [3].
3. Scalability is a good thing. The system can start small and then grow. For example, conveyor systems can increase from about 1,200 bags per hours to 1,500 bags per hour when needed [4]. Using variable speed drives also helps to save energy.

4. Another issue is that 5G in many developing airports is not available. So, it is safer to use local processing instead of depending on the internet all the time.

2.3 Environment

It is important to note that in airport terminals, where passengers and their luggage are checked in, about 75% of the airport's electricity costs [5]. The manufacturing of system components such as conveyor belts, motors, and sensors requires raw materials and energy, resulting in carbon emissions [6]. During operation, the system may contribute to environmental pollution through energy related emission and noise. At the end of its lifecycle, proper recycling and disposal of materials are necessary to reduce waste and environmental harm.

2.4 Economical

Aviation is one of the fastest-growing commercial sectors in the world right now, having over 4.5 billion passengers and 62 million tonnes of freight annually, which also contributes 3.6% of the world's GDP [7]. It is also important to note that baggage systems are expensive to install, but in the long run will save us loads of money as they are efficient. [8] While automation may reduce some manual jobs, it creates new opportunities in technical and system management roles.

2.5 Political

1. IATA Resolution 753 says airports must track luggage from start to finish, so this makes systems like ours very important [8]. But there is also a problem with data privacy laws like GDPR, because they need data to be encrypted, and this makes the system more expensive.

2. Government funding is not always stable in developing countries; sometimes, money can be cut. So, it is better to build the system in phases, not all at once, so if funding stops, the project can continue.

3. Safety rules are also very strict. The system must follow IEC standards; it will not be approved. Also, customers' rules are different in each country, so the system must be flexible to work with different security requirements.

3. Engineering Requirement Design Specifications

The Design Specification followed a structured approach where, firstly, we looked at the standards used, which were set by IATA for baggage dimensions and mass. We also looked at the performance requirements and found that the system's efficiency, uptime, tracking, and communication with passengers are important to consider to better show the system's capabilities. It was also important for us to better understand the power usage of this system, as energy is expensive and controlling how much power is used can help us save energy and money, which leads to things like maintenance and safety, which are priorities in this system (operation requirement).

Table 1: Design specifications

System efficiency $\geq 99.9\%$
System Uptime $\geq 99.5\%$ (24/7)
Tracking/detection Time ≤ 5 s under rated load
System throughput ≥ 1500 bags/hour
Baggage Dimension $200\text{mm} \leq \text{length} \leq 900\text{mm}$ $50\text{mm} \leq \text{width} \leq 750\text{mm}$ $200\text{mm} \leq \text{height} \leq 750\text{mm}$ [8]
Baggage Mass: Normal operating range: $2 \text{ kg} \leq \text{Mass} \leq 32 \text{ kg}$ [8]. Bags above 32 kg must be sent to manual overweight handling and must not enter the automatic sorter.
Communication with Passenger: <60 s upon arrival time
Cost < 4 Million
System Power Consumption $\leq 75\%$ of total Power
Repair time ≤ 30 minutes

4. Design Topic 4: Smart Airport Luggage Sorting and Tracking System Description:

BAGGAGE HANDLING

- (1) Transport bags from the check-in desks to the departure aircraft parking position [1]
- (2) transport bags from one aircraft parking position to another during transfers, and transport bags from the arrival parking position to the baggage-reclaim area. [1]

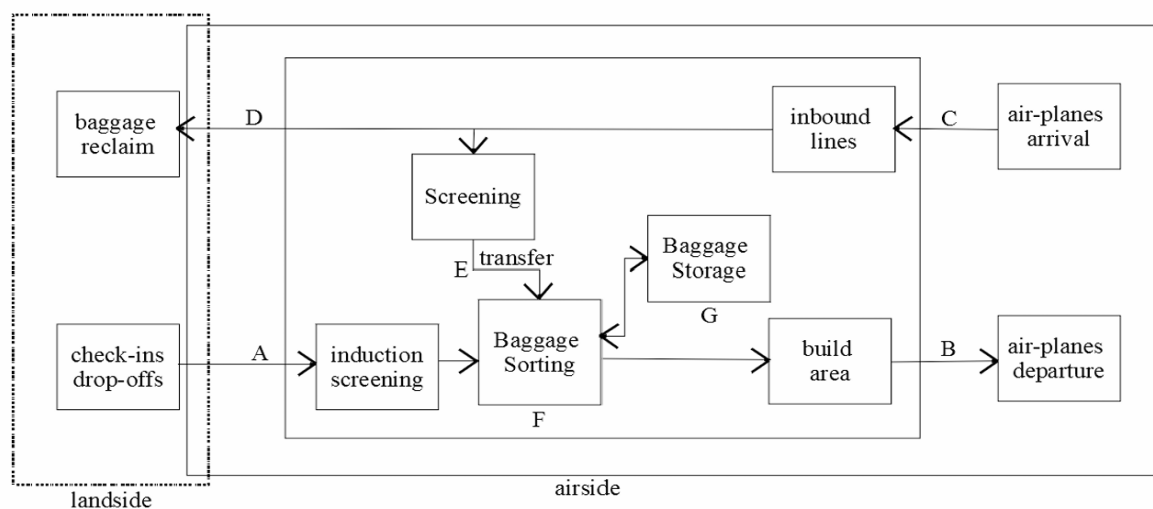


Figure 1: Flow diagram of BHS (Baggage Handling System)

Conveyors

Sorting system

A mechanical system that sorts bags in the correct direction and is made possible thanks to the barcode and the IT systems and controls. An IT system and PLC are used to automate the mechanical sorting system, which is the most suitable destination for a given bag.

5. Solutions

5.1 Alternative Solution 1

Carousel Sorter

In a baggage handling system, check-in bags are first identified and security screened, then are sent to a sorter. The sorter is a mechanical system with multiple chutes, each using a pusher controller by a local panel to direct bags to the correct destination. Sensors are used to ensure safe operation, and proximity switches monitor the pushers'

positions, and a photocell checks that each bag has been successfully moved and that the path is clear for the next one. The system can run at about 1 m/s and handle up to 2700 bags per hour[1].

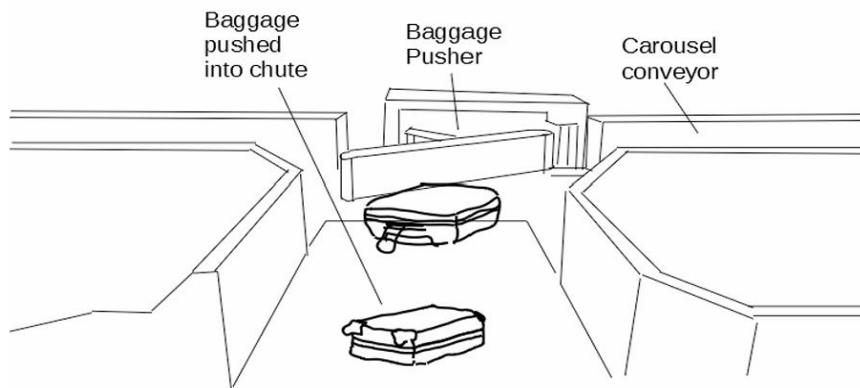


Figure 2: Carousel sorter

Motor drive specification

- Single main motor (unlike cross-belt)
- Moderate torque, steady operation

Microcontroller /Control system

- PLC
- Simple embedded controller for small systems

Model type

Rule-based: Carousel systems are simple and repetitive, making rule-based control sufficient and more reliable

5.2 Alternative Solution 2

Cross-Belt Sorter

In a baggage handling system, a cross-belt sorter is built on a continuous loop conveyor, and it uses many small conveyor belts mounted across the track. Each small belt moves independently and transfers bags sideways to the correct chute, because bags are carried and gently placed rather than pushed, handling is smoother, and reduces damage or jams. Depending on the system design, it can handle up to about 25 000 units per hour [1].

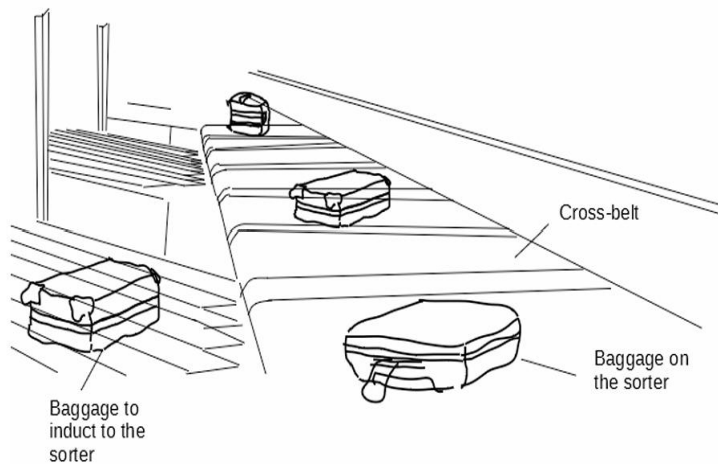


Figure 3: Cross-Belt Sorter

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Motor drive specification

- Multiple small motors (one per carrier)
- High-efficiency motors (often brushes or AC motors)
- Encoder-feedback for precise positioning

Microcontroller /Control system

- PLC
- Raspberry Pi or Arduino (for prototype)

Model type

Rule-based: The cross-belt sorter requires real-time, deterministic control, making rule-based logic good for sorting operations.

5.3. Tracking, Communication, and Passenger Interface

Both sorting systems will be using Radio Frequency Identification-Based Tracking with Short Message Service Passenger Notifications

5.3.1 System Description

This solution proposes installing Radio Frequency Identification portal readers at fixed checkpoints throughout the Airport baggage handling area. Each piece of luggage is

fitted with a passive Radio Frequency Identification (RFID) tag at check-in. When the bag passes through each checkpoint portal, the reader will automatically scan the tag and update the bag's status in the admin database. Passengers are notified of important status changes via automated Short Message Service messages sent to their registered mobile numbers. Airports handle thousands of bags daily, and this solution provides a reliable, low-maintenance, and cost-effective way to track every bag from check-in to the arrival carousel.

5.3.2 Operation

- At check-in, a passive Radio Frequency Identification tag is printed and attached to each bag. The tag stores a unique bag identifier linked to the passenger's flight details in the database.
- As the bag moves along the conveyor, Radio Frequency Identification portal readers installed at seven checkpoint zones automatically scan the tag without any manual intervention.
- Each successful scan updates the bag's status in the database in real time.
- The notification service detects each status change and sends an automated Short Message Service message to the passenger's mobile number.
- If a bag misses an expected checkpoint scan within a set time window, the system raises an alert on the staff dashboard for immediate investigation.
- At the arrival's carousel, a final scan confirms the bag has been placed for collection, and a Short Message Service message is sent to the passenger.

5.3.3 Major Components

- Ultra High Frequency Radio Frequency Identification tags
- Radio Frequency Identification portal readers at seven checkpoint zones
- Central SQL database server
- Short Message Service notification gateway (Bulk SMS)
- Staff operations dashboard
- Local area network and cabling infrastructure

5.3.4 Installation

- Radio Frequency Identification portal frames are installed at conveyor transition points, the security screening exit, make-up carousel entries, the aircraft loading dock, and the arrivals carousel.
- Ethernet cables connect all indoor readers to the central server room inside the terminal building.
- The central database server and notification gateway are hosted on a dedicated on-site server to comply with the Protection of Personal Information Act data residency requirements.
- The staff dashboard is accessed via standard web browsers on existing airport workstations with no additional hardware and software required.

5.3.5 Data Flow at the Airport

- Bag scan events travel from the Radio Frequency Identification reader to the central server over the airport's local area network.
- The server updates the bag status in the database and triggers the notification service.
- The notification service sends a Short Message Service message through the gateway to the passenger's phone within 15 seconds of the scan event.
- Staff supervisors monitor all seven checkpoint zones through a dashboard, which refreshes every 10 seconds.
- Exception alerts appear on the dashboard when a bag has not been scanned at the next expected checkpoint within 8 minutes.

5.3.6 Comparing Other Technologies

Table 2: Comparing Other Technologies

Technology	Range	Cost	Need for Internet	Works in developing countries?
GSM	Nationwide	Low	No	Yes
RFID	A few meters	Very Low	No	Yes
GPS	Global	Medium	No	Yes
Wi-Fi/Bluetooth	Short range	Low	Yes	Limited
IoT	Global	Medium	Yes	Depends on the internet

6. Recommend one of the alternatives and justify your choice

The recommendation followed a structured Weighted Objective or Multi-Criteria Decision analysis, where we first looked at important factors needed for the system to work well. Cost is very important because money is limited in developing countries, so the system must be cheap and can be done in phases. We also checked the technical part, such as whether the system is easy to use and easy to fix. We look at how many bags it can handle so it matches the airport's needs and avoids wasting money or causing delays. The passenger side is also important, so SMS must work on all phones, even without internet. Lastly, we look at risk when installing, because simple and tested systems are better in places with infrastructure challenges. All these things helped us choose a system that balances cost, performance, and easy use.

Table 3: Weight Objective

Criteria	Weight	Solution 1 Carousel Sort with (RFID+SMS)	Score	Weight Score 1	Solution 2 Cross-Belt Sorter with (RFID+SMS)	Score	Weight score 2
System Cost	0.2	Cheaper, simple system, uses a basic setup	5	1	Very expensive, complex system with many parts	2	0.4
Technical Complexity	0.15	Low, simple control system and easy to manage	3	0.45	High, need advanced control and more accuracy	5	0.75
Maintenance Requirements	0.2	Low, easy to fix, fewer parts, and no battery issues	5	1	Many motor parts need special care	2	0.4
Throughput Capacity	0.15	2,700 bags/hour is enough for the current need and	3	0.45	25,000 bags/hours , more reliable in big airports	4	0.6

		some growth					
Passenger Accessibility	0.15	Very good, SMS works for all phones	5	0.75	SMS works for all phones	5	0.75
Implementation Risk	0.15	Low risk, easy to install, and already proven	5	0.75	Higher risk, complex, and takes more time to set up	3	0.45
Total	1			4.4			3.35

Scoring scale: 1 to 5, as the score approaches 5, it becomes Excellent

Weight Score:

Sol 1: $(0.2 \times 5) + (0.15 \times 3) + (0.2 \times 5) + (0.15 \times 3) + (0.15 \times 5) + (0.15 \times 5)$
 $= 4.4$

Sol 2: $(0.2 \times 2) + (0.15 \times 5) + (0.2 \times 2) + (0.15 \times 4) + (0.15 \times 5) + (0.15 \times 3)$
 $= 3.35$

Score Breakdown

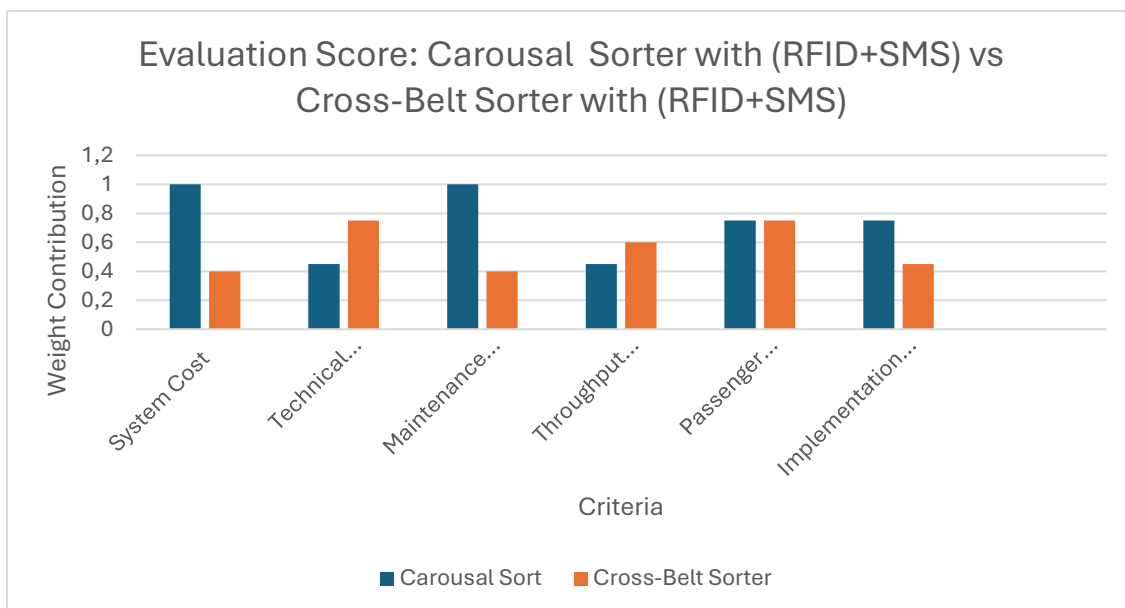


Figure 3 Comparison of Carousel Sorter with RFID+SMS vs Cross-Belt Sorter with RFID+SMS Tracking

Based on the calculation results above, the Carousel Sorter with RFID+SMS Tracking System is the recommended solution to be implemented.

6.1 This decision is backed by the following justifications:

- The Carousel Sorter is better because it is much cheaper and simpler to use [9]. This may be very important in developing countries where funding is not always stable, so projects must be done step by step [10]. It can still meet the needed requirements for baggage tracking without spending too much money.
 - It also has low maintenance because it uses one main motor and fewer parts, so it is easy to fix and does not need special tools or skills [9]. This helps the airport fix problems quickly. The Cross-Belt system has many small motors and is more complicated, so it needs more maintenance and expert knowledge [9].
 - The Carousel Sorter also has a low risk because it is already proven and easy to install [9]. This is good for airports where power and internet may not always be stable [10]. It can work reliably without too many problems during setup.
 - Its speed of about 2,700 bags per hour is enough for current needs and allows some future growth [9]. The Cross-Belt system is much faster, but that extra speed is not really needed now, and makes it more expensive [9].
 - Both systems use SMS, so passengers can receive updates on any phone, even basic ones without internet [10]. This is very important in developing countries.
- The Carousel Sorter is also simple in design, using basic PLC control, which makes it easier for workers and students to understand and manage [9].
- Even though the Cross-Belt Sorter is more advanced and faster, it is too expensive, complex, and harder to maintain. So, the Carousel Sorter is the better and more practical choice for now [9] [10].

7. Case studies:

In O.R. Tambo International Airport, advanced technologies have been implemented to reduce errors in luggage handling and improve operational efficiency. The airport introduced an automated baggage handling system integrated with barcode scanning and radio-frequency identification (RFID) tracking to ensure each bag is accurately identified, sorted, and routed in real time. This system is supported by centralized control software that monitors bag movement across conveyors and sorters, significantly reducing manual intervention and the risk of human error. In addition, redundancy mechanisms and real-time fault detection were incorporated following previous system disruptions, enabling quicker identification and resolution of faults before they escalate. These improvements have enhanced baggage tracking accuracy, reduced the incidence of lost or misrouted luggage, and improved overall passenger

satisfaction, demonstrating the effectiveness of automation, digital tracking, and system monitoring in modern airport baggage handling operations [11], [12].

7.1 RFID -based baggage tracking designed and implemented in Africa (Nigeria)

A Nigerian design team developed an RDIF-based luggage tracking system over 14 weeks to replace error-prone barcode methods, which aimed at increasing the success rates to nearly 100%. The solution showed that it mounted scanners for automated routing and a mobile application providing passengers with real-time baggage updates [13].

8. CONCLUSION

In conclusion, this project successfully designed a smart airport luggage sorting and tracking system that improves efficiency and reduces baggage handling errors. The comparison between the carousel sorter and cross-belt sorter showed that the carousel sorter with RFID and SMS integration is the most suitable solution for developing environments due to its simplicity, lower cost, and reliability. The use of RFID tracking and real-time SMS notifications enhances baggage visibility and minimizes the risk of lost and misrouted luggage. Also, the system design considers practical challenges such as infrastructure limitations, maintenance, and scalability. All in all, the proposed system provides a cost-effective and efficient solution that can significantly improve airport baggage handling operations.

8.1 Division of work

Nthethelelo	Andisa	Bhongo
Introduction	Technology Solution	STEEP (Social, Environmental, and Economic)
Abstract	Case study 1	Case study 2
STEEP ANALYSIS (Technical and Political)	Conclusion	Engineering requirements
Recommended solution	Recommended solution	Sorting solutions
References		References

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